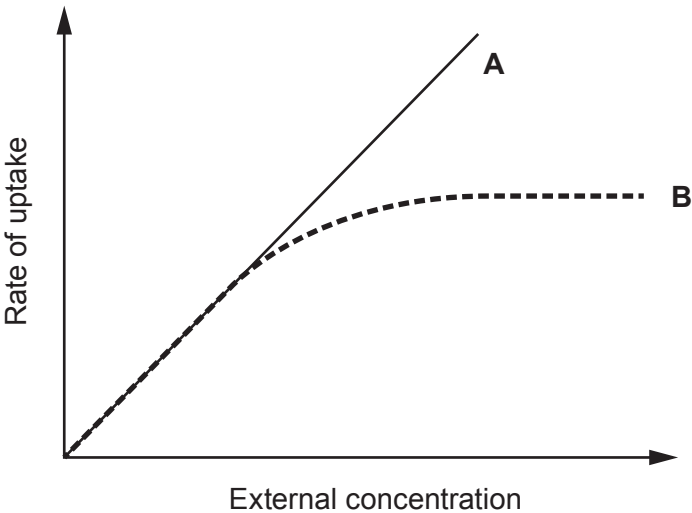


3. (a) Red blood cells need to take in glucose and oxygen from the blood plasma. Glucose is a polar molecule and oxygen is non-polar.

The uptake of glucose and oxygen by red blood cells was measured separately at increasing external concentrations of these two molecules.

Graph 3.1 shows the rate of uptake of glucose and oxygen.

Graph 3.1



- (i) **Complete Table 3.2** to identify:
- which line on **Graph 3.1** represents glucose and which represents oxygen;
 - the type of transport used by each substance.
- [2]

Table 3.2

Line	Substance	Type of transport
A		
B		



Examiner
only

(ii) Explain the difference in the shape of lines **A** and **B** shown on **Graph 3.1**. [4]

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(b) Red blood cells are packed full of haemoglobin.
When mature, they do not contain any of the organelles usually found in eukaryotic cells.

Using this information and your knowledge of **cell structure** conclude why mature red blood cells:

(i) cannot make haemoglobin; [2]

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(ii) can only transport substances across the cell membrane against a concentration gradient at very low rates. [2]

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